

✓ Merge Operations in Pandas

✓ We can merge dataframes using the following methods

- `pd.concat( [df] , axis)`
- `pd.merge( df1, df2 ,on,how,suffixes)`

More detailed documentation here : https://pandas.pydata.org/pandas-docs/stable/user_guide/merging.html

```
import pandas as pd
import numpy as np
```

```
imdb_df=pd.read_csv('imdb_data.csv')
```

```
imdb_df.head()
```



	id	belongs_to_collection		budget	genres	homepage	imdb_id	original_language	original_title	overview
0	1	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...		14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	tt2637294	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...
1	2	[[{'id': 107674, 'name': 'The Princess Diaries ...		40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'Romance'}]]	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...
2	3		NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under the direction of a ruthless instructor, ...
3	4		NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'name': 'Drama'}]]	http://kahaanithefilm.com/	tt1821480	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...
4	5		NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'name': 'Thriller'}]]	NaN	tt1380152	ko	마린보이	Marine Boy is the story of a former national s...

5 rows × 23 columns



```
imdb_df.drop('id',axis=1,inplace=True)
```

✓ Let us go ahead and create three subsets of our original dataset

```
# Create three dataframes
imdb1 = imdb_df[imdb_df['runtime'] >= 150]
imdb2 = imdb_df[ (imdb_df['runtime'] < 150) & (imdb_df['runtime'] > 100 )]
imdb3 = imdb_df[imdb_df['runtime'] <= 100]
```

```
imdb1.head()
```

	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview	popula
47	NaN	44000000	[[{'id': 80, 'name': 'Crime'}, {'id': 18, 'name': 'Drama'}]]	http://thehatefuleight.com/	tt3460252	en	The Hateful Eight	Bounty hunters seek shelter from a raging bliz...	20.32
51	NaN	12000000	[[{'id': 12, 'name': 'Adventure'}, {'id': 35, 'name': 'Comedy'}]]	NaN	tt0059243	en	The Great Race	Professional daredevil and white-suited hero, ...	4.47
76	NaN	500000	[[{'id': 18, 'name': 'Drama'}, {'id': 10749, 'name': 'Family'}]]	NaN	tt0242572	ta	Kandukondain Kandukondain	Two young women with differing views on love f...	0.66
97	NaN	0	[[{'id': 14, 'name': 'Fantasy'}, {'id': 18, 'name': 'Drama'}]]	NaN	tt1772980	ru	Mishen	In the year 2020, a group of wealthy Moscovite...	1.04
107	NaN	72000000	[[{'id': 28, 'name': 'Action'}, {'id': 18, 'name': 'Drama'}]]	NaN	tt0112573	en	Braveheart	Enraged at the slaughter of Murron, his new br...	20.75

5 rows × 22 columns

imdb2.head()

	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview	pop
1	[[{'id': 107674, 'name': 'The Princess Diaries ...'}]]	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'Drama'}]]	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	1.0
2	NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under the direction of a ruthless instructor, ...	6.0
3	NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'name': 'Drama'}]]	http://kahaanithefilm.com/	tt1821480	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...	1.0
4	NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'name': 'Thriller'}]]	NaN	tt1380152	ko	마린보이	Marine Boy is the story of a former national s...	1.0
10	[[{'id': 1575, 'name': 'Rocky Collection', 'pos...'}]]	1000000	[[{'id': 18, 'name': 'Drama'}]]	NaN	tt0075148	en	Rocky	When world heavyweight boxing champion, Apollo...	1.0

5 rows × 22 columns

```
imdb3.head()
```

	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview
0	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	tt2637294	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...
5	NaN	8000000	[[{'id': 16, 'name': 'Animation'}, {'id': 12, 'name': 'Comedy'}]]	NaN	tt0093743	en	Pinocchio and the Emperor of the Night	Pinocchio and his friends, a glow worm and a m...
6	NaN	14000000	[[{'id': 27, 'name': 'Horror'}, {'id': 53, 'name': 'Thriller'}]]	http://www.thepossessionmovie.com/	tt0431021	en	The Possession	A young girl buys an antique box at a yard sal...
7	NaN	0	[[{'id': 99, 'name': 'Documentary'}]]	NaN	tt0391024	en	Control Room	A chronicle which provides a rare window into ...
8	[[{'id': 256377, 'name': 'The Muppet Collection...]	0	[[{'id': 28, 'name': 'Action'}, {'id': 35, 'name': 'Comedy'}]]	NaN	tt0117110	en	Muppet Treasure Island	After telling the story of Flint's last journe...

5 rows × 22 columns

```
# Concat both axis
imdb_non_null_runtime = pd.concat([imdb1,imdb2,imdb3])
```

```
imdb_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 22 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   belongs_to_collection                 604 non-null   object
1   budget                               3000 non-null  int64
2   genres                               2993 non-null  object
3   homepage                             946 non-null   object
4   imdb_id                              3000 non-null  object
5   original_language                     3000 non-null  object
6   original_title                       3000 non-null  object
7   overview                             2992 non-null  object
8   popularity                           3000 non-null  float64
9   poster_path                          2999 non-null  object
10  production_companies                  2844 non-null  object
11  production_countries                  2945 non-null  object
12  release_date                         3000 non-null  object
13  runtime                              2998 non-null  float64
14  spoken_languages                     2980 non-null  object
15  status                               3000 non-null  object
16  tagline                              2403 non-null  object
17  title                                3000 non-null  object
18  Keywords                             2724 non-null  object
19  cast                                 2987 non-null  object
20  crew                                 2984 non-null  object
21  revenue                              3000 non-null  int64
dtypes: float64(2), int64(2), object(18)
memory usage: 515.8+ KB
```

```
imdb_df.shape
```

↩ (3000, 22)

imdb1.shape,imdb2.shape,imdb3.shape

↩ ((135, 22), (1586, 22), (1277, 22))

```
list_of_dfs = []
df = pd.read_csv("imdb_data.csv")
list_of_dfs.append(df)
```

```
consolidated_df = pd.concat(list_of_dfs)
```

Start coding or [generate](#) with AI.

```
# Concatenate along a particular axis
sample_df = pd.concat([imdb1, imdb2],axis=1)
```

```
sample_df.tail()
```

↩

	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview	popularity	poster_path	...
2994	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	..
2995	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	..
2996	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	..
2997	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	..
2999	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	..

5 rows × 44 columns

```
sample_df.info()
```

↩

```
<class 'pandas.core.frame.DataFrame'>
Index: 1721 entries, 47 to 2999
Data columns (total 44 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   belongs_to_collection                21 non-null     object
1   budget                              135 non-null    float64
2   genres                              134 non-null    object
3   homepage                            40 non-null     object
4   imdb_id                             135 non-null    object
5   original_language                    135 non-null    object
6   original_title                       135 non-null    object
7   overview                            135 non-null    object
8   popularity                          135 non-null    float64
9   poster_path                         135 non-null    object
10  production_companies                 125 non-null    object
11  production_countries                 134 non-null    object
12  release_date                         135 non-null    object
```

```

13 runtime                135 non-null    float64
14 spoken_languages       135 non-null    object
15 status                 135 non-null    object
16 tagline                85 non-null     object
17 title                 135 non-null    object
18 Keywords               120 non-null    object
19 cast                   132 non-null    object
20 crew                   132 non-null    object
21 revenue                135 non-null    float64
22 belongs_to_collection  297 non-null    object
23 budget                 1586 non-null   float64
24 genres                 1584 non-null   object
25 homepage               529 non-null    object
26 imdb_id                1586 non-null   object
27 original_language      1586 non-null   object
28 original_title         1586 non-null   object
29 overview               1586 non-null   object
30 popularity              1586 non-null   float64
31 poster_path            1586 non-null   object
32 production_companies    1535 non-null   object
33 production_countries    1568 non-null   object
34 release_date           1586 non-null   object
35 runtime                1586 non-null   float64
36 spoken_languages       1580 non-null   object
37 status                 1586 non-null   object
38 tagline                1312 non-null   object
39 title                 1586 non-null   object
40 Keywords               1476 non-null   object
41 cast                   1577 non-null   object
42 crew                   1576 non-null   object
43 revenue                1586 non-null   float64
dtypes: float64(8), object(36)
memory usage: 605.0+ KB

```

```
sample_df.columns
```

```

Index(['belongs_to_collection', 'budget', 'genres', 'homepage', 'imdb_id',
      'original_language', 'original_title', 'overview', 'popularity',
      'poster_path', 'production_companies', 'production_countries',
      'release_date', 'runtime', 'spoken_languages', 'status', 'tagline',
      'title', 'Keywords', 'cast', 'crew', 'revenue', 'belongs_to_collection',
      'budget', 'genres', 'homepage', 'imdb_id', 'original_language',
      'original_title', 'overview', 'popularity', 'poster_path',
      'production_companies', 'production_countries', 'release_date',
      'runtime', 'spoken_languages', 'status', 'tagline', 'title', 'Keywords',
      'cast', 'crew', 'revenue'],
      dtype='object')

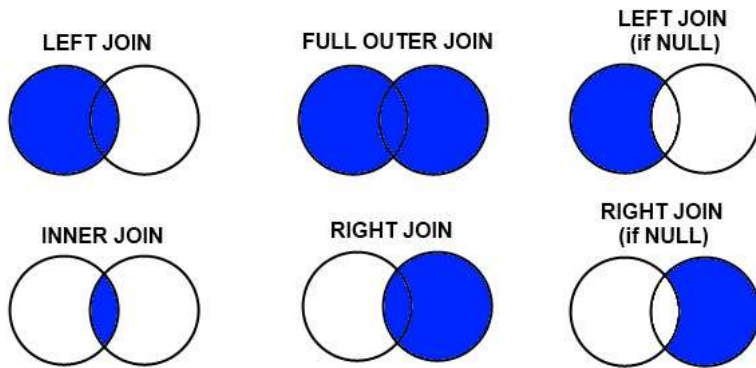
```

```
imdb1.shape, imdb2.shape
```

```
((135, 22), (1586, 22))
```

✓ Now let's do some joins

- Inner join
- Left join
- Right join
- Outer join



```
# Let us create a dataframe which consists of only English movies
english_movies = imdb_df[imdb_df['original_language'] == 'en']
```

```
# Next let us create another dataframe which consists of only those movies whose length is greater than 100 mins
long_movies = imdb_df.loc[imdb_df['runtime'] >= 100, ['imdb_id', 'original_language']]
```

```
english_movies.shape, long_movies.shape
```

```
((2575, 22), (1820, 2))
```

```
type(long_movies)
```

```
pandas.core.frame.DataFrame
def __init__(data=None, index: Axes | None=None, columns: Axes | None=None, dtype: Dtype | None=None, copy: bool | None=None) -> None
DataFrame.from_records : Constructor from tuples, also record arrays.
DataFrame.from_dict : From dicts of Series, arrays, or dicts.
read_csv : Read a comma-separated values (csv) file into DataFrame.
read_table : Read general delimited file into DataFrame.
read_clipboard : Read text from clipboard into DataFrame.
```

✓ Using inner join to merge the two dataframes

```
# Merging the two dataframes using inner join
merged_IJ_df = pd.merge(english_movies, long_movies, how='inner', left_on='imdb_id', right_on='imdb_id')
```

```
merged_IJ_df
```

	belongs_to_collection	budget	genres	homepage	imdb_id	original_language_x	original_title	overview
0	[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'Drama'}]]	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Thermop is no colle gradu and
1	NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under director a ruthless instructor
2	[[{'id': 256377, 'name': 'The Muppet Collection...	0	[[{'id': 28, 'name': 'Action'}, {'id': 35, 'name': 'Adventure'}]]	NaN	tt0117110	en	Muppet Treasure Island	After tell the story Flint's i journ
3	[[{'id': 1575, 'name': 'Rocky Collection', 'pos...	1000000	[[{'id': 18, 'name': 'Drama'}]]	NaN	tt0075148	en	Rocky	When we heavywei box champi Apoll
4	NaN	15000000	[[{'id': 18, 'name': 'Drama'}]]	http://www.dreamworks.com/ab/	tt0169547	en	American Beauty	Les Burnhan depres: suburt father i
...	...	...	...	...	...	...	...	...
1523	NaN	60000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'Drama'}]]	http://www.theterminal-themovie.com/	tt0362227	en	The Terminal	Vil Navorsk a n withou country; l
1524	NaN	0	[[{'id': 18, 'name': 'Drama'}]]	NaN	tt0105327	en	School Ties	When De Gree receive footi scholar
1525	NaN	0	[[{'id': 35, 'name': 'Comedy'}, {'id': 10749, 'name': 'Adventure'}]]	NaN	tt0109403	en	Chasers	Military n Rock Re and Ed Devane
1526	NaN	65000000	[[{'id': 80, 'name': 'Crime'}, {'id': 28, 'name': 'Drama'}]]	NaN	tt0116908	en	The Long Kiss Goodnight	Saman Cai suburt homelal is the id
1527	NaN	35000000	[[{'id': 53, 'name': 'Thriller'}, {'id': 28, 'name': 'Drama'}]]	http://www.abductionthefilm.com/	tt1600195	en	Abduction	A you man s ou uncover truth abo

```
temp_df = imdb_df[(imdb_df['original_language'] == 'en') & (imdb_df['runtime'] >=100)]
```

```
temp_df.shape
```

```
(1528, 22)
```

✦ Using left join to merge the two dataframes

```
# Merging the two dataframes using left join
merged_LJ_df = pd.merge(english_movies, long_movies, how='left',left_on='imdb_id',right_on='imdb_id')
```

merged_LJ_df

✓ Using right join to merge the two dataframes

```
# Merging the two dataframes using right join
merged_RJ_df = pd.merge(english_movies, long_movies, how='right',left_on=['imdb_id'],right_on='imdb_id')
```

merged_RJ_df

	belongs_to_collection	budget	genres	homepage	imdb_id	original_language_x	original_title	over
0	{'id': 107674, 'name': 'The Princess Diaries ...	40000000.0	{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Thermc is n cc grac a
1	NaN	3300000.0	{'id': 18, 'name': 'Drama']}	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under directi a ruti instruct
2	NaN	NaN	NaN	NaN	tt1821480	NaN	NaN	
3	NaN	NaN	NaN	NaN	tt1380152	NaN	NaN	
4	{'id': 256377, 'name': 'The Muppet Collection...	0.0	{'id': 28, 'name': 'Action'}, {'id': 35, 'nam...	NaN	tt0117110	en	Muppet Treasure Island	After t the stc Flint' jou
...	...	...	...	...	...	...	...	
1815	NaN	0.0	{'id': 18, 'name': 'Drama']}	NaN	tt0105327	en	School Ties	When [Gr receiv for schol
1816	NaN	0.0	{'id': 35, 'name': 'Comedy'}, {'id': 10749, '...	NaN	tt0109403	en	Chasers	Military Rock I and E Devan
1817	NaN	NaN	NaN	NaN	tt2364975	NaN	NaN	
1818	NaN	65000000.0	{'id': 80, 'name': 'Crime'}, {'id': 28, 'name...	NaN	tt0116908	en	The Long Kiss Goodnight	Sam C sub homem is the
1819	NaN	35000000.0	{'id': 53, 'name': 'Thriller'}, {'id': 28, 'n...	http://www.abductionthefilm.com/	tt1600195	en	Abduction	A y man c uncover truth at

1820 rows × 23 columns

✓ Using outer join to merge the two dataframes

```
# Outer Join
merged_OJ_df = pd.merge(english_movies[['imdb_id','runtime']],long_movies, on='imdb_id', how='outer')
```